Stock and purchase estimate

Revision bookcase-baseline
Spec SHA256 2d4c00c1478a2cc839f2c3a32bbd6d2072196de2e0f8cb736ab08d9d3560aa2b

Layout: complete. Deterministic first-fit rows; optimality is not claimed.

Saw kerf: 3.2 mm. Edge trim: 10 mm.

Per-edge rough allowance: 0 mm; finish to nominal size separately.

Grain-locked boards follow the declared compatible stock grain direction.

Area lower bounds do not establish sufficient purchasing quantities.

Sheet purchase costs (user-entered USD prices)

main_sheet: 1 sheet(s), unit USD unknown, purchased USD unknown

back_sheet: 1 sheet(s), unit USD unknown, purchased USD unknown

Known sheet subtotal: USD 0.00

Sheet subtotal: incomplete; missing price or layout

Complete project total: unknown. Excludes hardware, finish, tax, shipping, tools.

Consumed allocation: finished face area / original sheet face area. Currency: HALF_EVEN to cents.

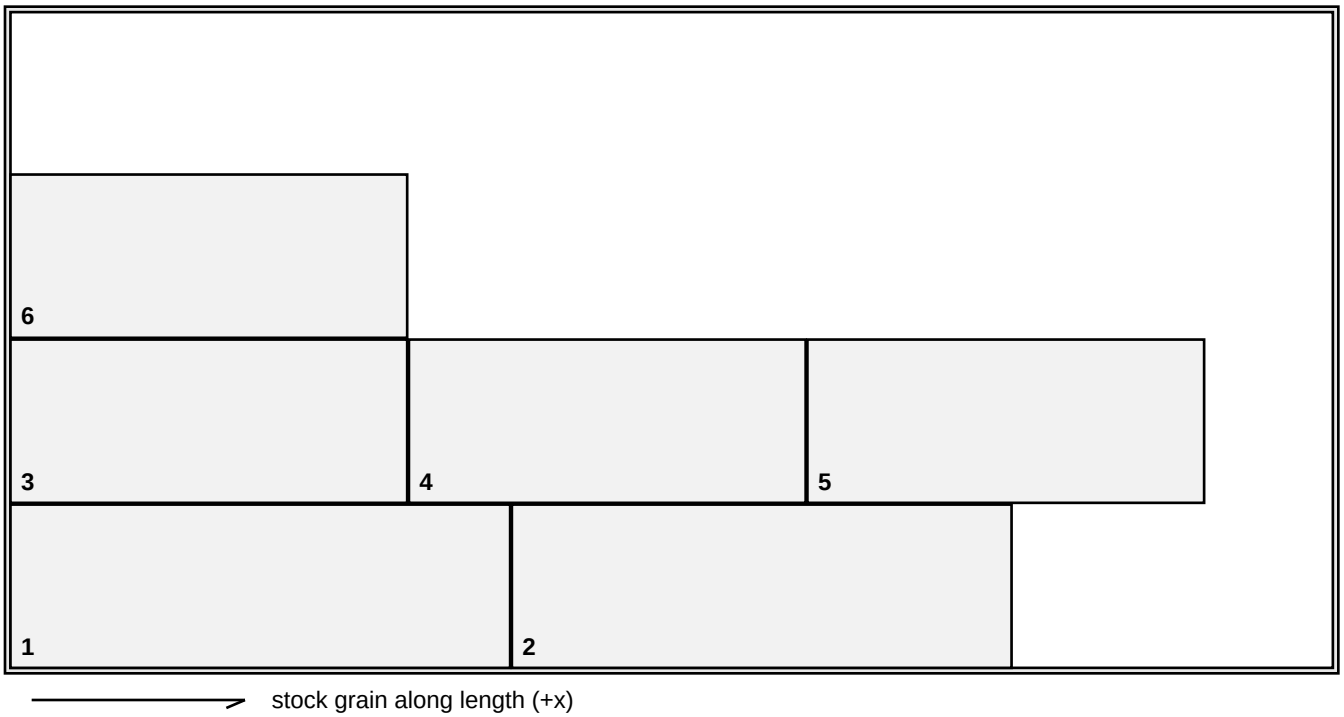
main_sheet / sheet 1

Revision bookcase-baseline

Spec SHA256 2d4c00c1478a2cc839f2c3a32bbd6d2072196de2e0f8cb736ab08d9d3560aa2b

Original sheet: approx. 96 1/16 in (2440 mm) x approx. 48 1/32 in (1220 mm)

Material plywood_main; actual thickness 18 mm; grain length



Part IDs and rough rectangles; coordinates from the original sheet corner (mm)

1. side_left: x=10, y=10, 914.4 x 298.8, rotation 0 deg
2. side_right: x=927.6, y=10, 914.4 x 298.8, rotation 0 deg
3. bottom: x=10, y=312, 726 x 298.8, rotation 0 deg
4. shelf_01: x=739.2, y=312, 726 x 298.8, rotation 0 deg
5. shelf_02: x=1468.4, y=312, 726 x 298.8, rotation 0 deg
6. top: x=10, y=614, 726 x 298.8, rotation 0 deg

Areas mm2: parts 1414160.64; internal kerf 28968.96; offcuts 1460870.4

Usable 2904000; reserved edge bands 72800; original 2976800

Cut sequence: main_sheet / 1

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Spec SHA256 2d4c00c1478a2cc839f2c3a32bbd6d2072196de2e0f8cb736ab08d9d3560aa2b

Apply reserved edge trim first, then replay each numbered internal cut on its named region.

Internal kerf removes material. No trailing kerf is invented at a finished stock boundary.

trim_01: trim region original along x; kerf begins 6.8 mm, width 3.2 mm.
Source x=0, y=0, 2440 x 1220 mm. Result: discard_left / after_left.

trim_02: trim region after_left along x; kerf begins 2430 mm, width 3.2 mm.
Source x=10, y=0, 2430 x 1220 mm. Result: trimmed_x / discard_right.

trim_03: trim region trimmed_x along y; kerf begins 6.8 mm, width 3.2 mm.
Source x=10, y=0, 2420 x 1220 mm. Result: discard_front / after_front.

trim_04: trim region after_front along y; kerf begins 1210 mm, width 3.2 mm.
Source x=10, y=10, 2420 x 1210 mm. Result: usable / discard_back.

cut_01: region usable, axis y, kerf begins 308.8 mm, width 3.2 mm
Kerf begins 298.8 mm from this active region origin along the named axis.
Source x=10, y=10, 2420 x 1200 mm; children row_1 / remainder_1

cut_02: region remainder_1, axis y, kerf begins 610.8 mm, width 3.2 mm
Kerf begins 298.8 mm from this active region origin along the named axis.
Source x=10, y=312, 2420 x 898 mm; children row_2 / remainder_2

cut_03: region remainder_2, axis y, kerf begins 912.8 mm, width 3.2 mm
Kerf begins 298.8 mm from this active region origin along the named axis.
Source x=10, y=614, 2420 x 596 mm; children row_3 / offcut_top

cut_04: region row_1, axis x, kerf begins 924.4 mm, width 3.2 mm
Kerf begins 914.4 mm from this active region origin along the named axis.
Source x=10, y=10, 2420 x 298.8 mm; children side_left / row_1_remaining

cut_05: region row_1_remaining, axis x, kerf begins 1842 mm, width 3.2 mm
Kerf begins 914.4 mm from this active region origin along the named axis.
Source x=927.6, y=10, 1502.4 x 298.8 mm; children side_right / offcut_row_1

cut_06: region row_2, axis x, kerf begins 736 mm, width 3.2 mm
Kerf begins 726 mm from this active region origin along the named axis.
Source x=10, y=312, 2420 x 298.8 mm; children bottom / row_2_remaining_a

cut_07: region row_2_remaining_a, axis x, kerf begins 1465.2 mm, width 3.2 mm
Kerf begins 726 mm from this active region origin along the named axis.
Source x=739.2, y=312, 1690.8 x 298.8 mm; children shelf_01 / row_2_remaining_b

Cut sequence continued

Revision bookcase-baseline
Spec SHA256 2d4c00c1478a2cc839f2c3a32bbd6d2072196de2e0f8cb736ab08d9d3560aa2b

cut_08: region row_2_remaining_b, axis x, kerf begins 2194.4 mm, width 3.2 mm

Kerf begins 726 mm from this active region origin along the named axis.

Source x=1468.4, y=312, 961.6 x 298.8 mm; children shelf_02 / offcut_row_2

cut_09: region row_3, axis x, kerf begins 736 mm, width 3.2 mm

Kerf begins 726 mm from this active region origin along the named axis.

Source x=10, y=614, 2420 x 298.8 mm; children top / offcut_row_3

These diagrams describe sheet cutting; they are not a drill-access or structural approval.

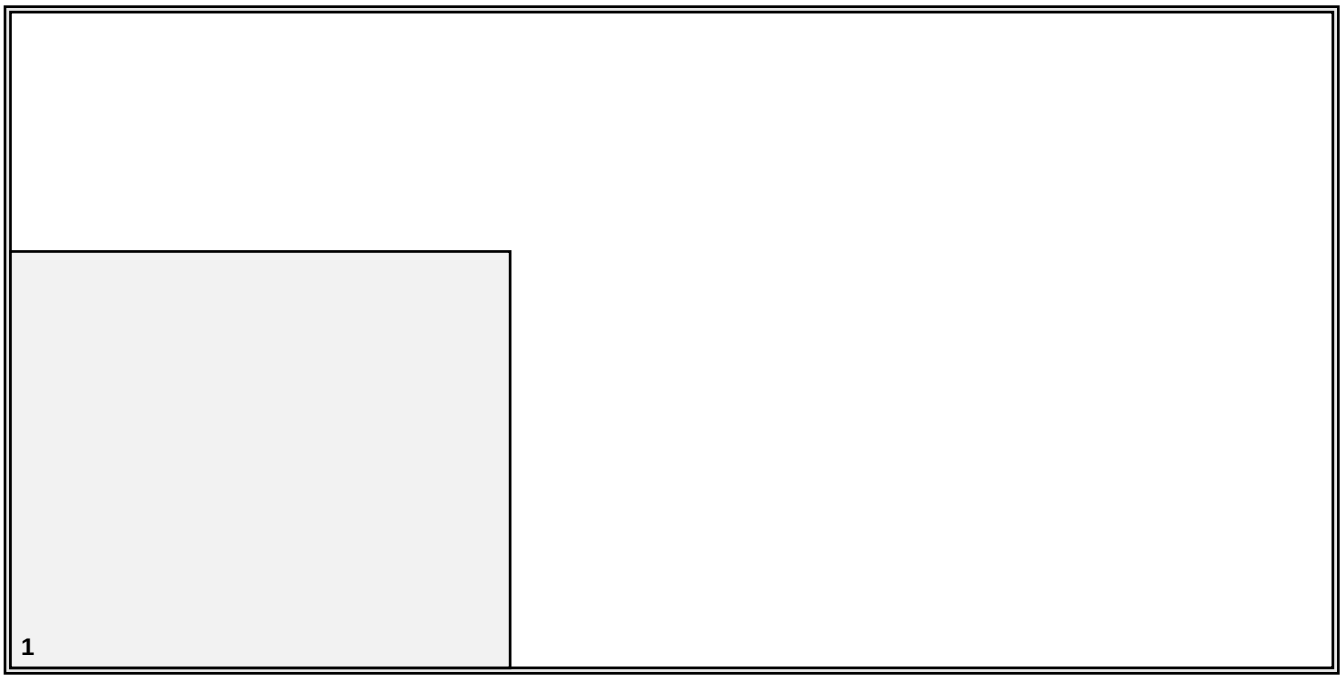
back_sheet / sheet 1

Revision bookcase-baseline

Spec SHA256 2d4c00c1478a2cc839f2c3a32bbd6d2072196de2e0f8cb736ab08d9d3560aa2b

Original sheet: approx. 96 1/16 in (2440 mm) x approx. 48 1/32 in (1220 mm)

Material plywood_back; actual thickness 6 mm; grain length



Part IDs and rough rectangles; coordinates from the original sheet corner (mm)

1. back: x=10, y=10, 914.4 x 762, rotation 0 deg

Areas mm2: parts 696772.8; internal kerf 10182.4; offcuts 2197044.8

Usable 2904000; reserved edge bands 72800; original 2976800

Cut sequence: back_sheet / 1

Revision bookcase-baseline
Spec SHA256 2d4c00c1478a2cc839f2c3a32bbd6d2072196de2e0f8cb736ab08d9d3560aa2b

Apply reserved edge trim first, then replay each numbered internal cut on its named region.

Internal kerf removes material. No trailing kerf is invented at a finished stock boundary.

trim_01: trim region original along x; kerf begins 6.8 mm, width 3.2 mm.

Source x=0, y=0, 2440 x 1220 mm. Result: discard_left / after_left.

trim_02: trim region after_left along x; kerf begins 2430 mm, width 3.2 mm.

Source x=10, y=0, 2430 x 1220 mm. Result: trimmed_x / discard_right.

trim_03: trim region trimmed_x along y; kerf begins 6.8 mm, width 3.2 mm.

Source x=10, y=0, 2420 x 1220 mm. Result: discard_front / after_front.

trim_04: trim region after_front along y; kerf begins 1210 mm, width 3.2 mm.

Source x=10, y=10, 2420 x 1210 mm. Result: usable / discard_back.

cut_01: region usable, axis y, kerf begins 772 mm, width 3.2 mm

Kerf begins 762 mm from this active region origin along the named axis.

Source x=10, y=10, 2420 x 1200 mm; children back_row / offcut_top

cut_02: region back_row, axis x, kerf begins 924.4 mm, width 3.2 mm

Kerf begins 914.4 mm from this active region origin along the named axis.

Source x=10, y=10, 2420 x 762 mm; children back / offcut_right

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